Dark Pattern Radar: Windows Setup

Run the app on your own Windows PC in about 5 minutes. No coding needed.

What you received

A zip file called **DarkPatternRadar.zip**. Inside is a small web app that checks whether shopping-site text is a 'dark pattern' (a pressure trick like a fake countdown). It runs **on your own computer** in your web browser. Nothing is uploaded anywhere.

You need: Windows 10 or 11, an internet connection for the first setup (to download packages), and about 500 MB of free space.

Step 1. Install Python (one time)

- Open **https://www.python.org/downloads/windows/** and download the latest **Python 3.12** (3.10, 3.11 or 3.13 also work) *Windows installer (64-bit)*.
- Run the installer. On the very first screen **tick the box 'Add python.exe to PATH'** at the bottom. This is the most common thing people miss.
- Click **Install Now** and wait until it says *Setup was successful*.
- Check it worked: press the Windows key, type **cmd**, press Enter. In the black window type the line below and press Enter. You should see a version number such as *Python 3.12.x*.

```
python --version
```

If it says *'python' is not recognized*, try **py --version**. If that works, use **py** instead of **python** in every command below. If neither works, reinstall Python and tick the PATH box.

Step 2. Unzip the app

- Save **DarkPatternRadar.zip** somewhere simple, e.g. your Downloads folder.
- Right-click it, choose **Extract All...**, then **Extract**. You get a folder called **DarkPatternRadar**.
- Tip: do not run the app from inside the zip. Always extract it first.

Step 3, option A (easiest): double-click the launcher

- Open the extracted **DarkPatternRadar** folder and double-click **run_app.bat**.
- If Windows shows 'Windows protected your PC', click **More info**, then **Run anyway** (it is a plain text script you can open in Notepad to read).
- The first run takes 2-5 minutes: it creates a private Python environment and installs the packages. Later runs start in seconds.
- Your browser opens at **http://localhost:8501**. If it does not, open that address yourself.

Step 3, option B: do it manually in Command Prompt

Use this if the launcher fails or you want to see what is happening. Replace **YourName** with your Windows user name; use the real folder path if you extracted elsewhere.

1. Open Command Prompt (Windows key, type **cmd**, Enter) and go to the app folder with **cd** (change directory):

```
cd C:\Users\YourName\Downloads\DarkPatternRadar
```

Tip: if the folder is on another drive (like D:), type the drive letter first: **D:** and press Enter, then use cd. Typing **dir** should list *app.py* and *requirements.txt*; if not, you are in the wrong folder.

2. Create a virtual environment (a private box for this app's packages, so nothing else on your PC is touched):

```
python -m venv venv
```

3. Activate it. The prompt will now start with **(venv)**:

```
venv\Scripts\activate
```

4. Install the required packages (needs internet, takes 1-5 minutes):

```
python -m pip install --upgrade pip
pip install -r requirements.txt
```

This installs **streamlit** (the web app framework), **pandas** (data tables) and **scikit-learn** (the machine-learning model). Equivalent one-liner: **pip install streamlit pandas scikit-learn**.

5. Start the app:

```
streamlit run app.py
```

The first time, Streamlit may ask for an email address. Just **press Enter** to skip. Your browser then opens **http://localhost:8501**. If Windows Firewall asks, click **Allow** (only your own PC connects).

Using the app

- Click **Try example** or paste text from a shopping site into the box, then click **Analyze**.
- You get a verdict, a percentage (the model's probability that it is a dark pattern) and coloured chips showing which words pushed the answer: pink toward dark, green toward normal.
- Scroll down to **Dataset explorer** to search and filter the 2,356 real examples, and **Export CSV** to save the filtered rows.
- The first page load takes a few seconds while the model trains. After that it is instant.

Stopping and starting again

- **Stop:** click the Command Prompt window and press **Ctrl + C**, or just close the window.
- **Start again (launcher):** double-click run_app.bat.
- **Start again (manual):** open cmd, **cd** to the folder, run **venv\Scripts\activate**, then **streamlit run app.py**. You do not need to reinstall anything.

Troubleshooting

Problem	Fix
'python' is not recognized	Python is not on PATH. Try **py** instead of python. Otherwise rerun the Python installer, choose Modify or reinstall, and tick 'Add python.exe to PATH'. Close and reopen cmd afterwards.

Problem	Fix
'streamlit' is not recognized	The virtual environment is not active. Run **venv\Scripts\activate** first (prompt must start with (venv)), or use **python -m streamlit run app.py**.
pip install fails or times out	Check internet, then run the pip command again. Behind a school/work network try a phone hotspot. Make sure Python is 3.10 or newer.
Browser does not open / blank page	Open **http://localhost:8501** manually. Wait ~10 seconds on first load. Try another browser (Chrome, Edge).
'Port 8501 is already in use'	The app is already running in another window. Close it, or run **streamlit run app.py --server.port 8502** and open localhost:8502.
Error about missing data/dataset.tsv	You ran it from the wrong folder or from inside the zip. Extract the zip fully and **cd** into the DarkPatternRadar folder (the one containing app.py).
run_app.bat window closes instantly	Open cmd, cd to the folder, type **run_app.bat** to see the error message, or use option B.
Text looks unstyled / fonts look plain	The fonts load from Google Fonts. Without internet the app still works with fallback fonts.

What is inside the folder

File	What it is
app.py	the web app
data/dataset.tsv	real labeled dataset (yamanalab/ec-darkpattern, Apache-2.0)
requirements.txt	package list used by pip
run_app.bat	double-click launcher for Windows
pipeline.py	optional: prints model scores in the terminal (python pipeline.py)
Dark_Pattern_Audit_Model_Card.pdf	performance, bias audit and limitations